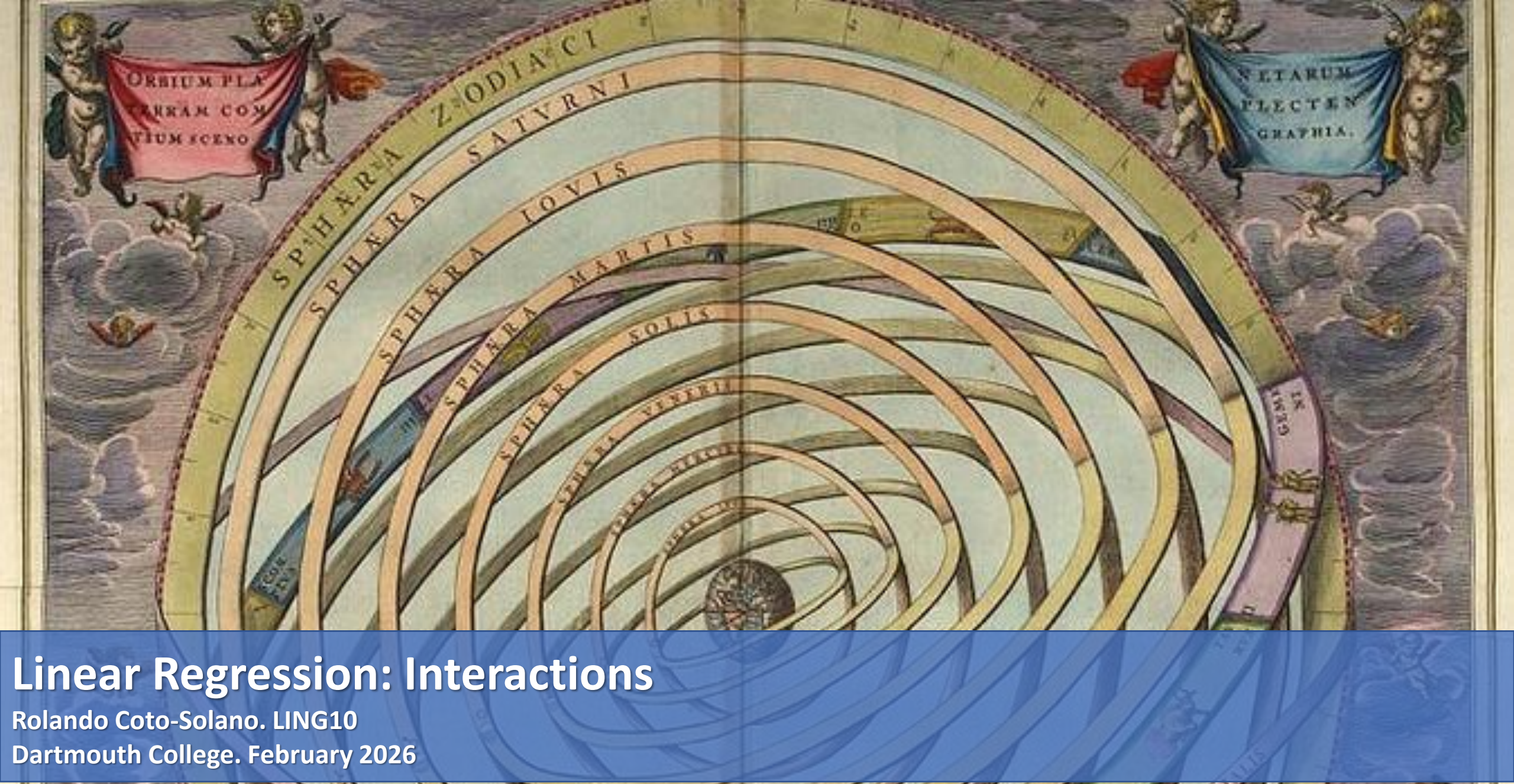


Linear Regression: Interactions

Rolando Coto-Solano. LING10
Dartmouth College. February 2026

Summary

This week we'll study linear regressions more in-depth:

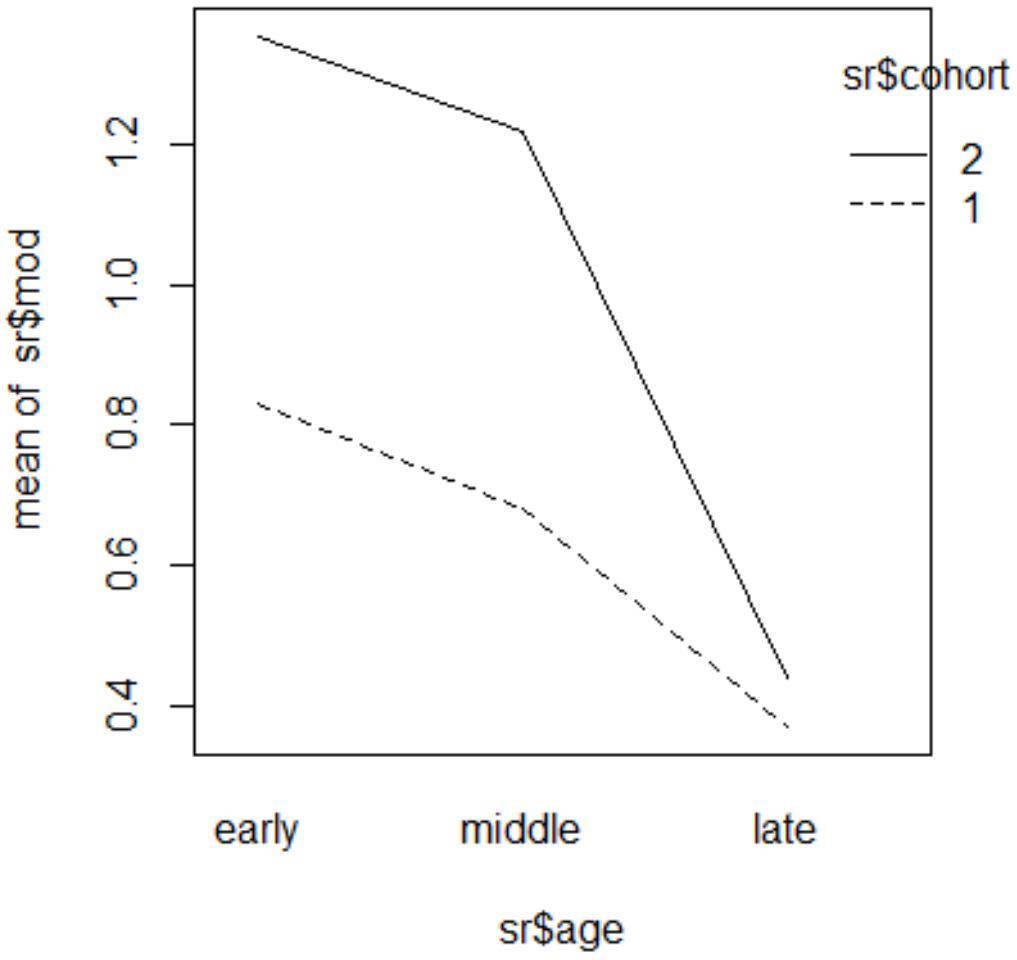
- Interactions
- Parsimony and variable selection

Interactions

An **interaction** between variables happens when the combination of the two changes their behavior.

The effect of one variable depends on a second variable. (The effects aren't additive).

Linear regressions can also study interactions between predictor variables.



Interactions: Nicaraguan Sign Language

Dependent (y) = Average of spatial modulation per verb (%, Numerical)

% Spatial modulations per verb
Spatial modulations: Signs are produced in a non-central position.
This indicates deictic and locative information.

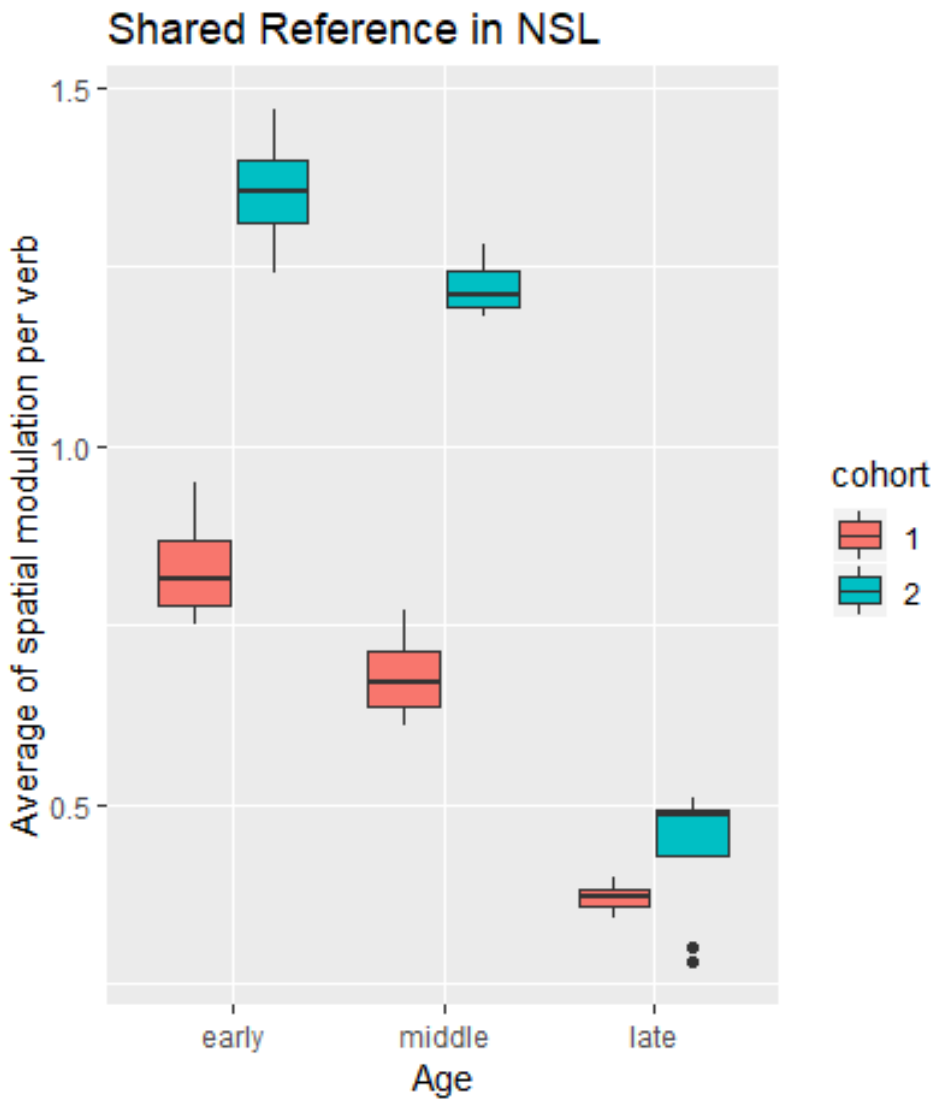
Independent 1 (x) = Age of children when they joined the school (Categorical)

Early (Before 6.5 years old)
Middle (Between 6.5 – 10 years old)
Late (After 10 years old)

Independent 2 (x) = Cohort: What decade did the person join:

1 and 2? (Early and middle?)

Interactions: Nicaraguan Sign Language



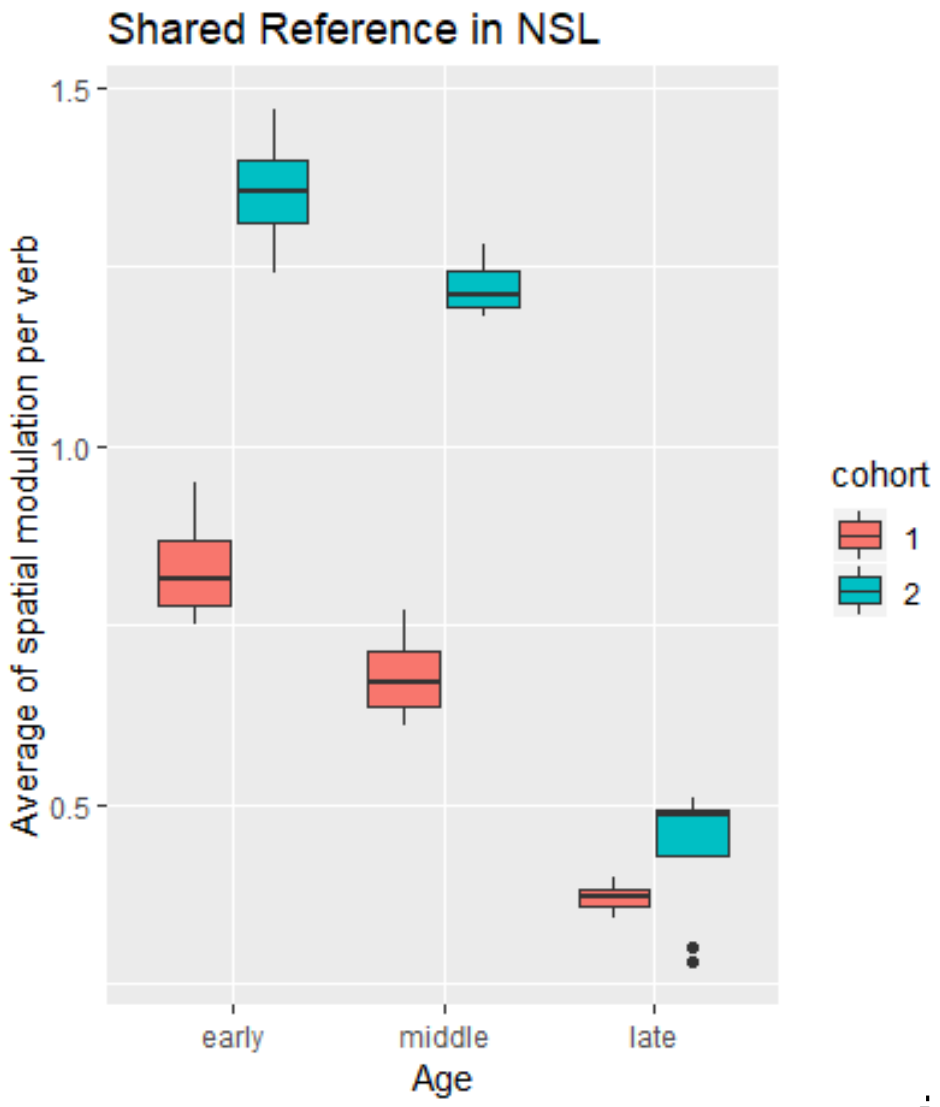
8 children per condition

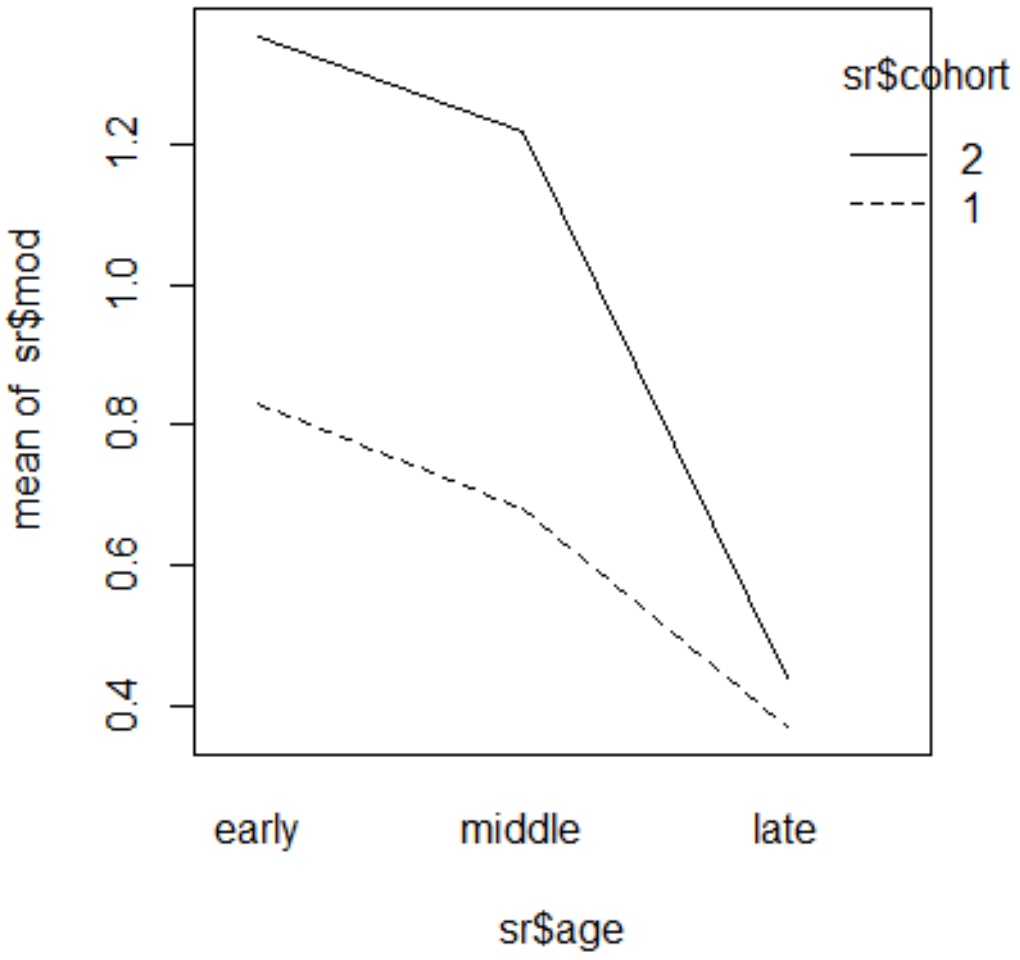
6 conditions (age,cohort):

Early-1
Middle-1
Late-1

Early-2
Middle-2
Late-2

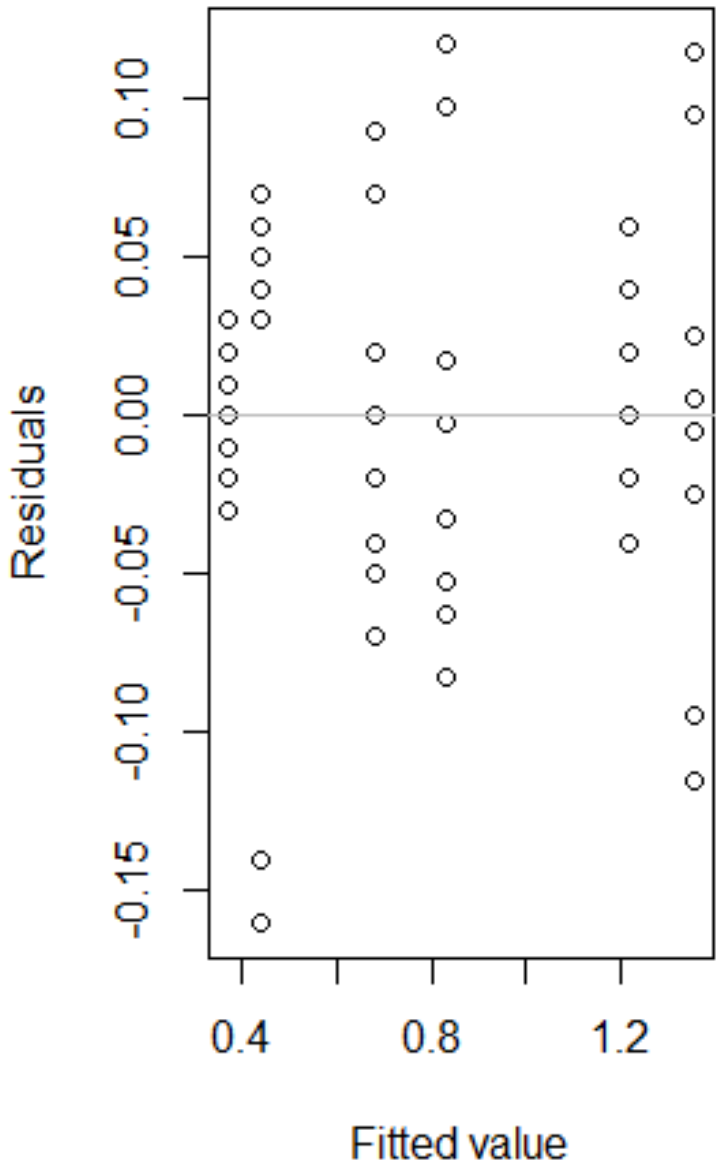
Interactions: Nicaraguan Sign Language

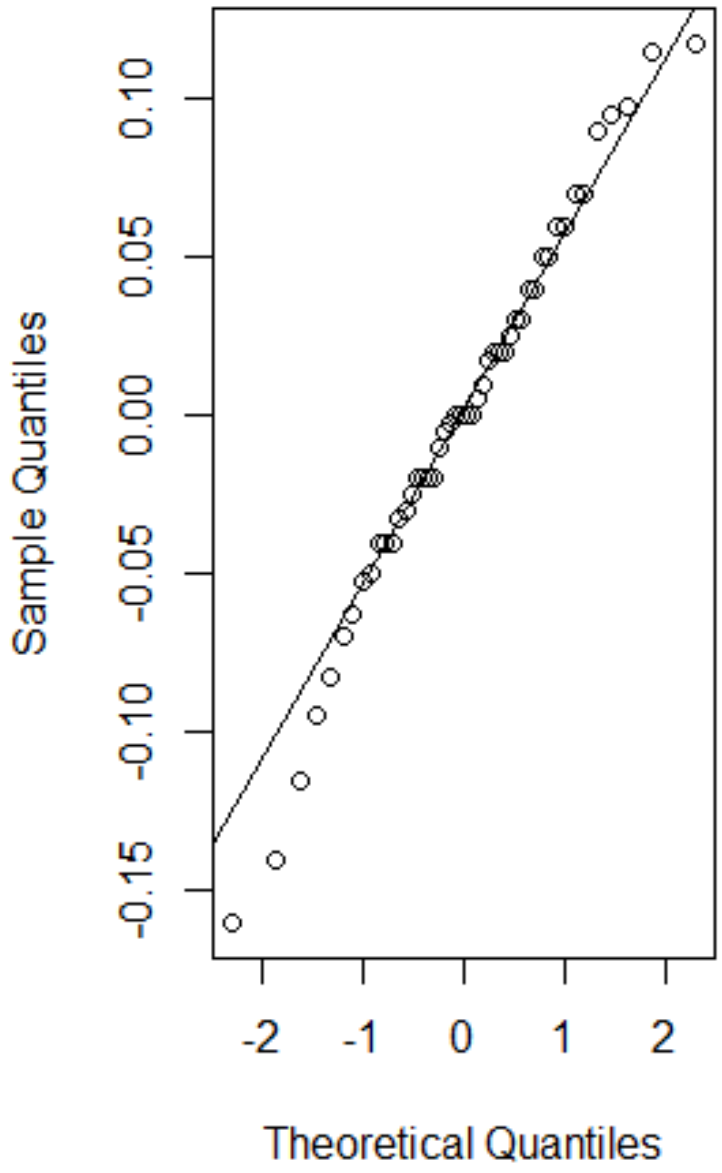





```
interaction.plot(sr$age, sr$cohort, sr$mod)
```

Interactions: Nicaraguan Sign Language





Interactions: Nicaraguan Sign Language

```
> shapiro.test(mNSL$residuals)
     Shapiro-Wilk normality test
data:  mNSL$residuals
W = 0.9824, p-value = 0.681

> ncvTest(mNSL)
Non-constant Variance Score Test
Variance formula: ~ fitted.values
Chisquare = 0.01741704, Df = 1, p = 0.89501

> durbinWatsonTest(mNSL)
 lag Autocorrelation D-W Statistic p-value
   1      -0.1231633      2.194726   0.948
 Alternative hypothesis: rho != 0
```

Interactions: Nicaraguan Sign Language

```
> vif(mNSL)
           GVIF Df GVIF^(1/(2*Df))
age           4  2        1.414214
cohort        3  1        1.732051
age:cohort    8  2        1.681793
```

Interactions

```
> mNSL = lm(mod ~ age*cohort, data = nsl)
> summary(mNSL)
Call:
lm(formula = mod ~ age * cohort, data = nsl)
Residuals:
     Min       1Q   Median       3Q      Max 
-0.16000 -0.03438  0.00000  0.04000  0.11750 
Coefficients:
                  Estimate Std. Error t value Pr(>|t|)    
(Intercept)        0.83250    0.02317  35.933  < 2e-16 ***
agemiddle         -0.15250    0.03276  -4.654 3.24e-05 ***
agelate           -0.46250    0.03276 -14.116  < 2e-16 ***
cohort2            0.52250    0.03276  15.947  < 2e-16 ***
agemiddle:cohort2  0.01750    0.04634   0.378    0.708    
agelate:cohort2   -0.45250    0.04634  -9.766 2.27e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 0.06553 on 42 degrees of freedom
Multiple R-squared:  0.973,	Adjusted R-squared:  0.9698 
F-statistic: 302.9 on 5 and 42 DF,  p-value: < 2.2e-16
```

Interactions

```
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)         0.83250    0.02317  35.933  < 2e-16 ***
agemiddle          -0.15250    0.03276  -4.654 3.24e-05 ***
agelate            -0.46250    0.03276 -14.116  < 2e-16 ***
cohort2             0.52250    0.03276  15.947  < 2e-16 ***
agemiddle:cohort2   0.01750    0.04634   0.378    0.708
agelate:cohort2    -0.45250    0.04634  -9.766 2.27e-12 ***
```

What is the equation?

Interactions

```
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)         0.83250    0.02317  35.933  < 2e-16 ***
agemiddle          -0.15250    0.03276  -4.654 3.24e-05 ***
agelate            -0.46250    0.03276 -14.116  < 2e-16 ***
cohort2             0.52250    0.03276  15.947  < 2e-16 ***
agemiddle:cohort2   0.01750    0.04634   0.378    0.708
agelate:cohort2    -0.45250    0.04634  -9.766 2.27e-12 ***
```

What is the equation?

mod = 0.83 -0.15(ageMiddle) – 0.46(ageLate) + 0.52(cohort2) +
0.017(ageMiddle*cohort2) – 0.45(ageLate*cohort2) + ε

Interactions

```
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)          0.83250    0.02317  35.933  < 2e-16 ***
agemiddle           -0.15250    0.03276  -4.654 3.24e-05 ***
agelate             -0.46250    0.03276 -14.116  < 2e-16 ***
cohort2              0.52250    0.03276  15.947  < 2e-16 ***
agemiddle:cohort2    0.01750    0.04634   0.378    0.708
agelate:cohort2     -0.45250    0.04634  -9.766 2.27e-12 ***
```

What is the equation?

mod = 0.83 -0.15(ageMiddle) - 0.46(ageLate) + 0.52(cohort2) +
0.017(ageMiddle*cohort2) – 0.45(ageLate*cohort2) + ε

$\beta_{ageMiddle}$: The difference between ageEarly and ageMiddle, for cohort1 (when cohort2=0).

Interactions

```
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)          0.83250    0.02317  35.933  < 2e-16 ***
agemiddle           -0.15250    0.03276  -4.654 3.24e-05 ***
agelate             -0.46250    0.03276 -14.116  < 2e-16 ***
cohort2              0.52250    0.03276  15.947  < 2e-16 ***
agemiddle:cohort2    0.01750    0.04634   0.378    0.708
agelate:cohort2     -0.45250    0.04634  -9.766 2.27e-12 ***
```

What is the equation?

mod = 0.83 -0.15(ageMiddle) – 0.46(ageLate) + 0.52(cohort2) + 0.017(ageMiddle*cohort2) – 0.45(ageLate*cohort2) + ε

$\beta_{cohort2}$: The difference between cohort1 and cohort2, for ageEarly (when ageMiddle=0 and ageLate=0)

Interactions

```
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)          0.83250    0.02317  35.933  < 2e-16 ***
agemiddle           -0.15250    0.03276  -4.654 3.24e-05 ***
agelate             -0.46250    0.03276 -14.116  < 2e-16 ***
cohort2              0.52250    0.03276  15.947  < 2e-16 ***
agemiddle:cohort2    0.01750    0.04634   0.378    0.708
agelate:cohort2     -0.45250    0.04634  -9.766 2.27e-12 ***
```

What is the equation?

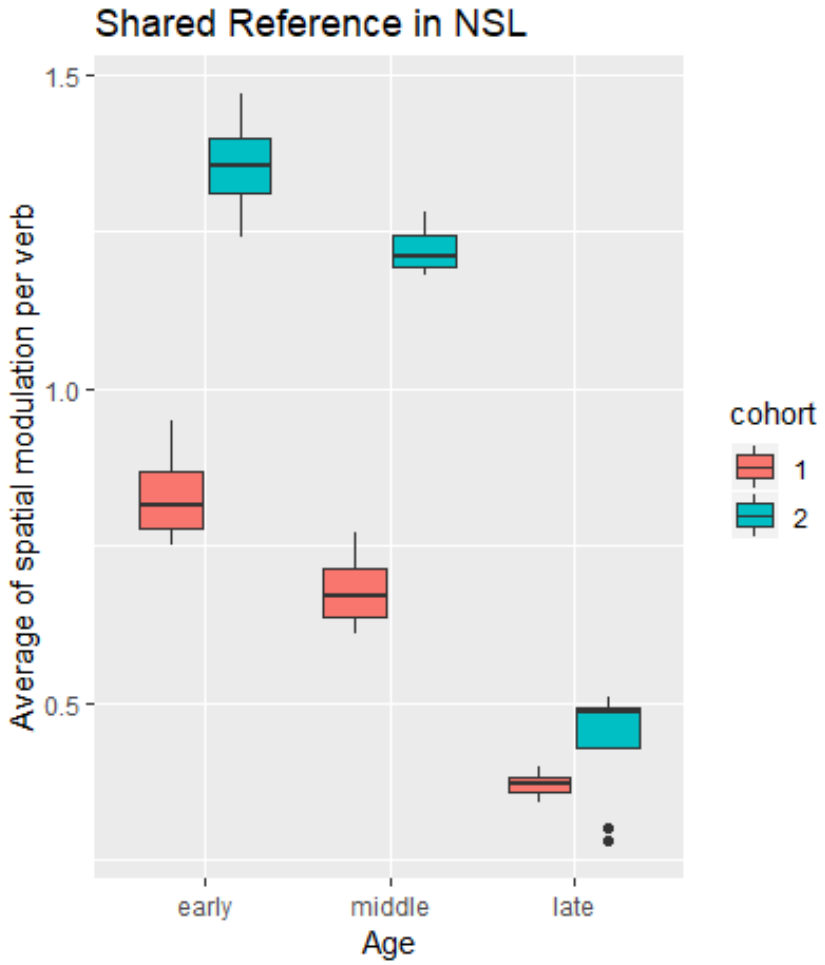
mod = 0.83 -0.15(ageMiddle) – 0.46(ageLate) + 0.52(cohort2) +
0.017(ageMiddle*cohort2) – 0.45(ageLate*cohort2) + ε

$\beta_{ageLate*cohort2}$: The amount of change in mod when ageLate=1 and cohort2=1
(when ageEarly = 0 and ageMiddle=0)

Interactions

mod = 0.83 -0.15(ageMiddle) – 0.46(ageLate) + 0.52(cohort2) +
0.017(ageMiddle*cohort2) – 0.45(ageLate*cohort2) + ε

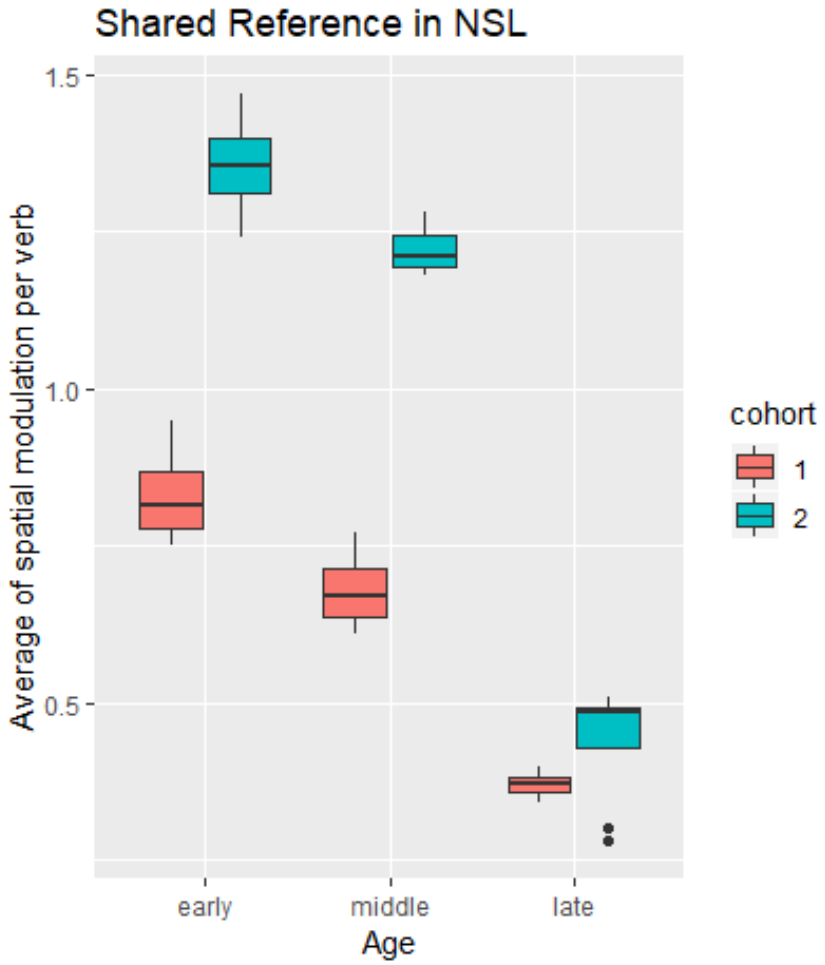
Age	Cohort	Prediction
Early	1	
Middle	1	
Late	1	
Early	2	
Middle	2	
Late	2	



Interactions

mod = 0.83 -0.15(ageMiddle) – 0.46(ageLate) + 0.52(cohort2) +
0.017(ageMiddle*cohort2) – 0.45(ageLate*cohort2) + ε

Age	Cohort	Prediction
Early	1	0.83
Middle	1	0.68
Late	1	0.37
Early	2	1.35
Middle	2	1.217
Late	2	0.44



Interactions

```
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)         0.83250    0.02317  35.933  < 2e-16 ***
agemiddle          -0.15250    0.03276  -4.654 3.24e-05 ***
agelate            -0.46250    0.03276 -14.116  < 2e-16 ***
cohort2             0.52250    0.03276  15.947  < 2e-16 ***
agemiddle:cohort2   0.01750    0.04634   0.378    0.708
agelate:cohort2    -0.45250    0.04634  -9.766 2.27e-12 ***
```

An **interaction** is a way to study two variables whose variation depends on one another. If your interaction is significant, explaining it is the most faithful explanation of the data.

Interactions

```
> lsmeans(mNSL, pairwise ~ age*cohort, adjust="tukey")
$contrasts
 contrast              estimate     SE df t.ratio p.value
 early,1 - middle,1      0.152 0.0328 42   4.654 0.0004
 early,1 - late,1        0.463 0.0328 42  14.116 <.0001
 early,1 - early,2      -0.522 0.0328 42 -15.947 <.0001
 early,1 - middle,2     -0.388 0.0328 42 -11.827 <.0001
 early,1 - late,2        0.393 0.0328 42  11.979 <.0001
 middle,1 - late,1       0.310 0.0328 42   9.461 <.0001
 middle,1 - early,2     -0.675 0.0328 42 -20.602 <.0001
 middle,1 - middle,2    -0.540 0.0328 42 -16.481 <.0001
 middle,1 - late,2       0.240 0.0328 42   7.325 <.0001
 late,1 - early,2       -0.985 0.0328 42 -30.063 <.0001
 late,1 - middle,2      -0.850 0.0328 42 -25.943 <.0001
 late,1 - late,2        -0.070 0.0328 42  -2.136 0.2890
 early,2 - middle,2      0.135 0.0328 42   4.120 0.0023
 early,2 - late,2        0.915 0.0328 42  27.927 <.0001
 middle,2 - late,2       0.780 0.0328 42  23.806 <.0001
P value adjustment: tukey method for comparing a family of 6 estimates
```

Interactions

```
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)         0.83250    0.02317  35.933  < 2e-16 ***
agemiddle          -0.15250    0.03276  -4.654 3.24e-05 ***
agelate            -0.46250    0.03276 -14.116  < 2e-16 ***
cohort2             0.52250    0.03276  15.947  < 2e-16 ***
agemiddle:cohort2   0.01750    0.04634   0.378    0.708
agelate:cohort2    -0.45250    0.04634  -9.766 2.27e-12 ***
```

There is a significant interaction between age and cohort ($t(42)=-9.8$, $p<0.0001$): Children in cohort 2 have more shared references than children in cohort 1 if they entered school at an early or middle age. However, children who entered school at a late age show no difference between the cohorts ($p=0.29$); and they have a lower rate than any of the other children.

Now you try it!

Exercise 1A: Dutch ont-

Dutch has a prefix *ont-* (e.g. *onthuld* ‘revealed’, *ontheven* ‘relieved’). Sometimes the *ont-* is followed by a plosive sound (e.g. *ontploft* ‘exploded’), and sometimes it’s not (e.g. *ontaarden* ‘to degenerate’). This information is in the column `PlosivePresent`. There is another column, `NumberOfSegmentsOnset`, which tells you the number of segments in the onset of the root syllable. (For example, e.g. *ontmoet* ‘met’ has 1 segment in the root onset, and *ontsnappen* ‘relax’ has two).

Do these two variables (whether a plosive is present or not, and the number of segments in the onset) influence the `DurationOfPrefix`?

Now you try it!

Exercise 1B:

F_0 of Turkish vowels and its interaction with the voicing of the preceding consonant.

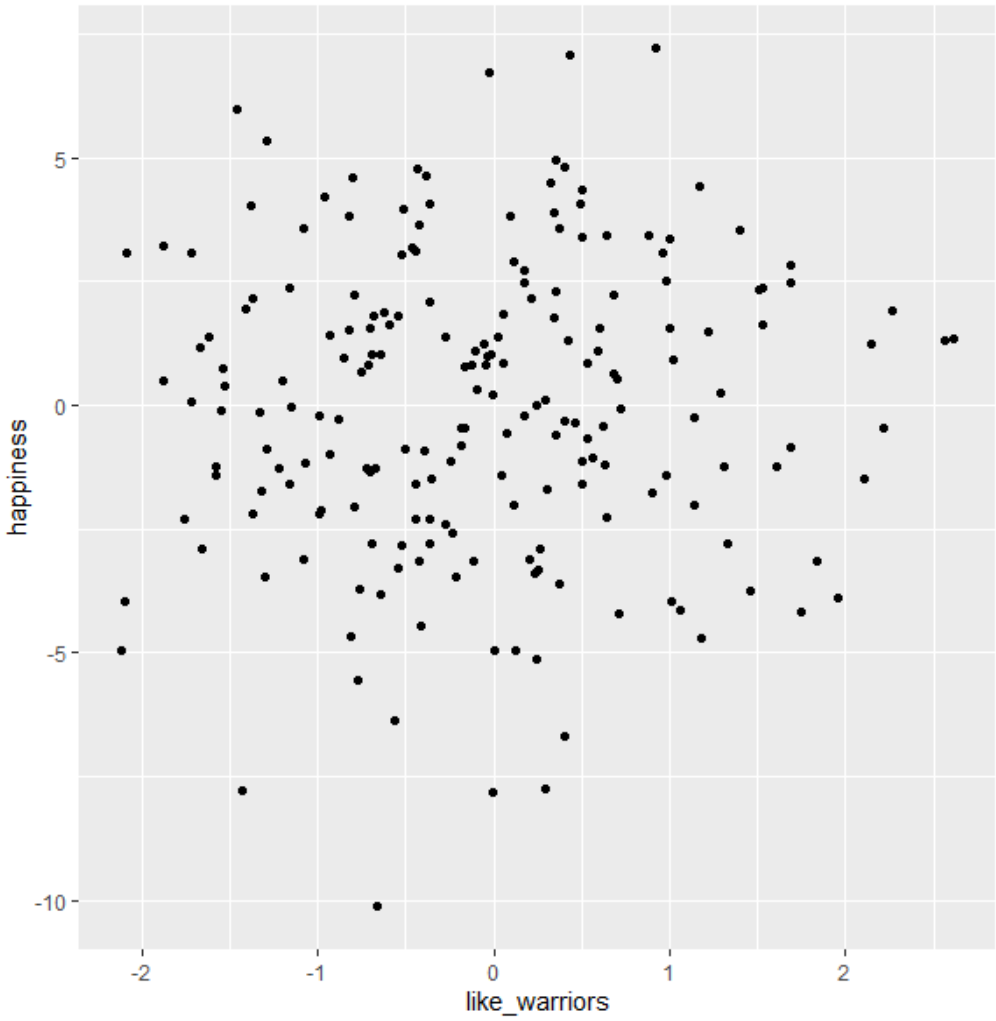
Numerical x Categorical Interactions

Let's use a non-linguistic example.

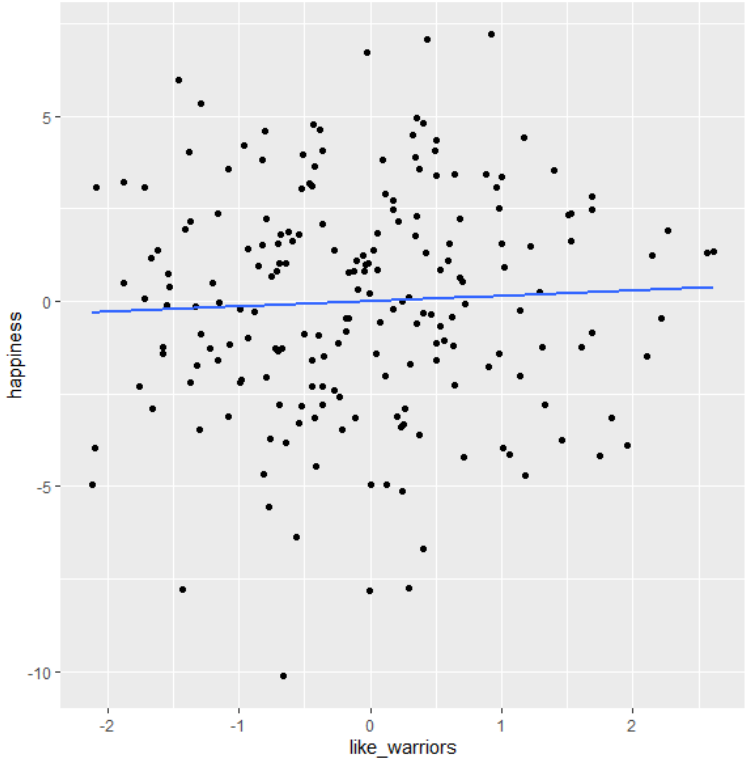
Dependent variable: `happiness` (continuous; how happy a person is [-11,7])

Independent variable: `like_warriors` (continuous, [-3,3])
`winner` (categorical: {cavs, warriors})

Numerical x Categorical Interactions

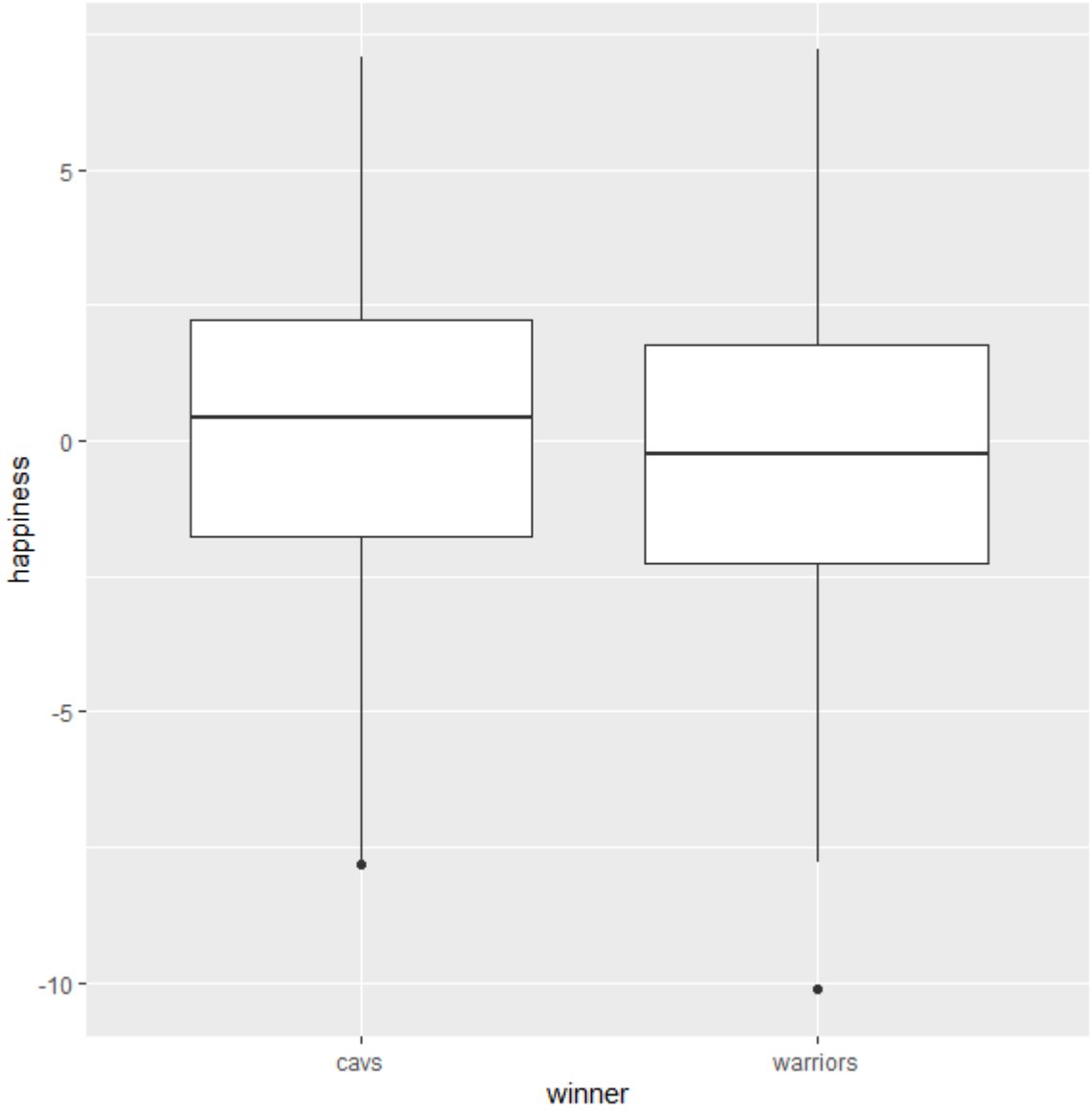


Numerical x Categorical Interactions

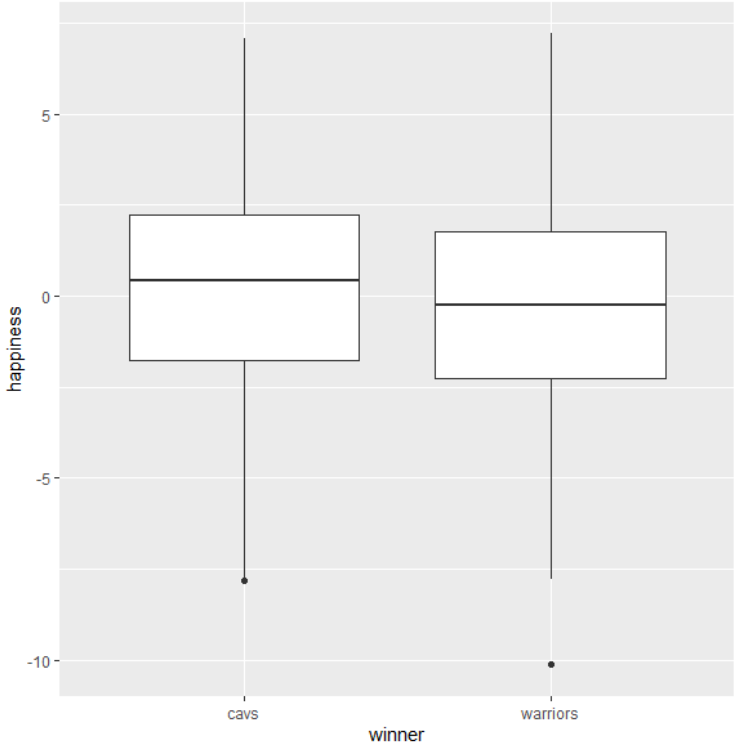



```
> m = lm(happiness ~ like_warriors, data)
> summary(m)

Call:
lm(formula = happiness ~ like_warriors, data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-10.0379  -1.9805   0.1235   2.1219   7.0763

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)   -0.005426   0.214761  -0.025    0.980
like_warriors  0.142651   0.214189   0.666    0.506

Residual standard error: 3.03 on 198 degrees of freedom
Multiple R-squared:  0.002235,    Adjusted R-squared:  -
0.002804
F-statistic: 0.4436 on 1 and 198 DF,  p-value: 0.5062
```

Numerical x Categorical Interactions

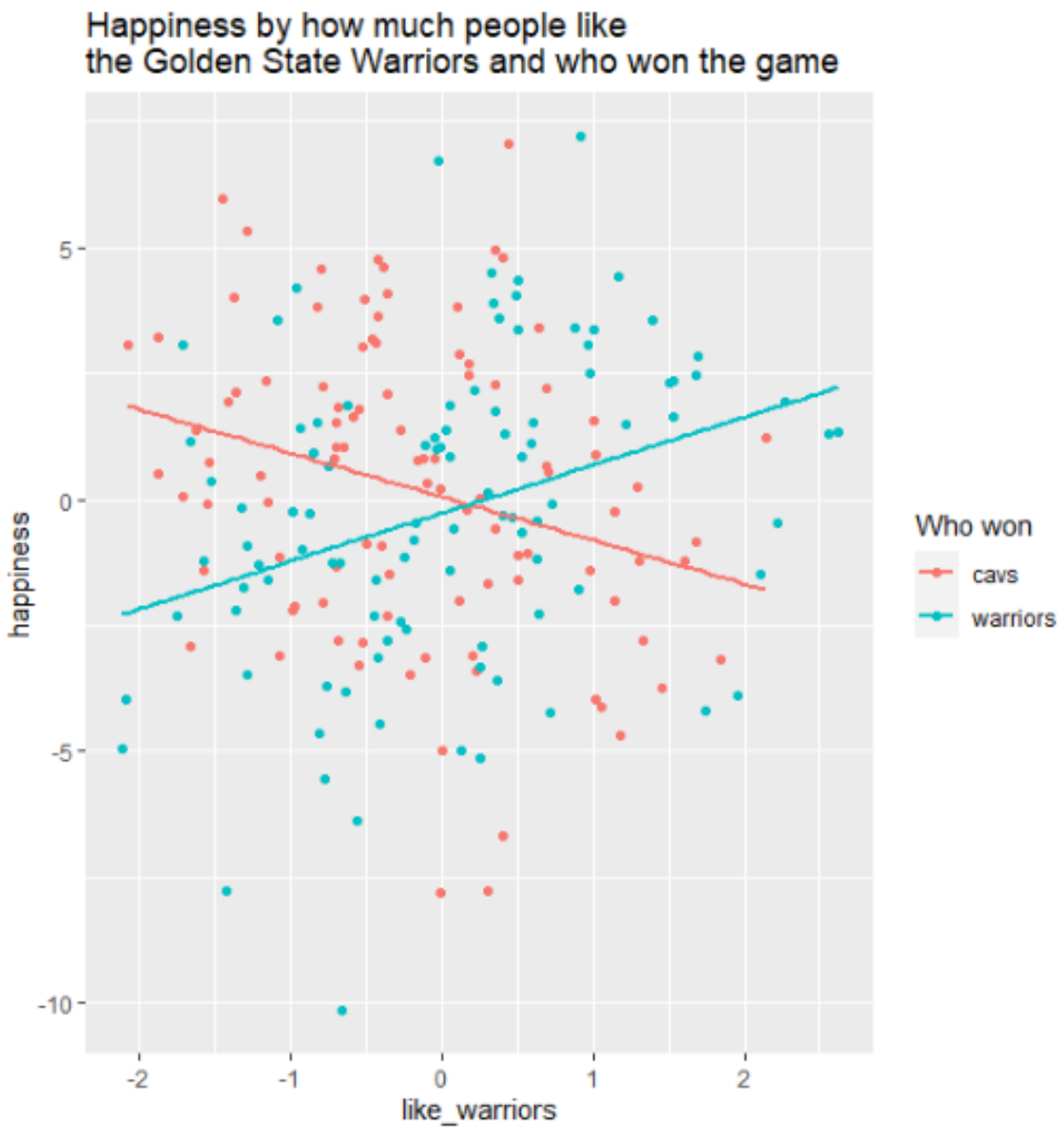


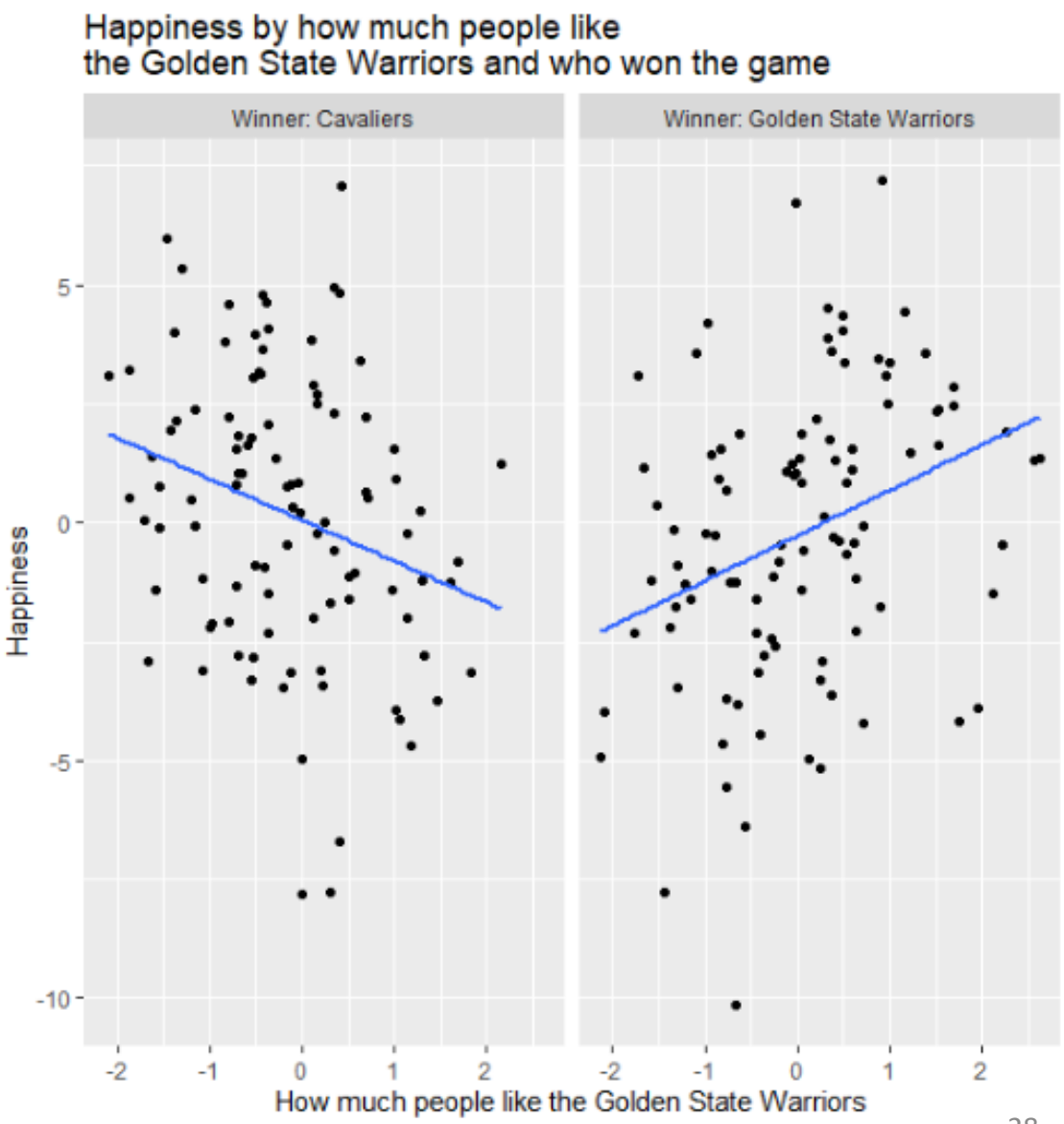
Numerical x Categorical Interactions



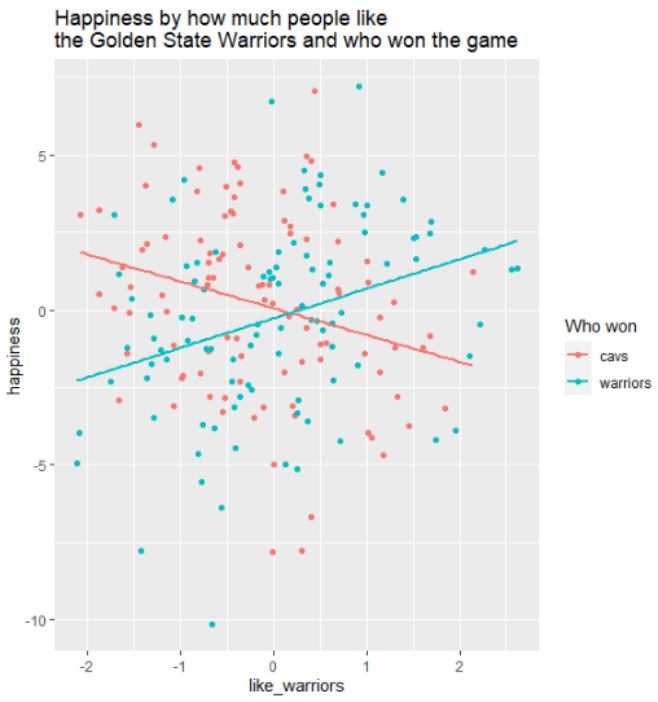

```
> m = lm(happiness ~ winner, data)
> summary(m)

Call:
lm(formula = happiness ~ winner, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-9.9052 -2.0473  0.0611  2.0373  7.4356 

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)      0.2033     0.3026   0.672    0.502
winnerwarriors  -0.4366     0.4279  -1.020    0.309

Residual standard error: 3.026 on 198 degrees of freedom
Multiple R-squared:  0.00523,	Adjusted R-squared:  
0.0002063 
F-statistic: 1.041 on 1 and 198 DF,  p-value: 0.3088
```

Numerical x Categorical Interactions





Numerical x Categorical Interactions




```
> m = lm(happiness ~ winner*like_warriors, data)
> summary(m)

Call:
lm(formula = happiness ~ winner * like_warriors, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-9.2337 -1.9096  0.1452  1.7537  7.3910 

Coefficients:
                            Estimate Std. Error t value Pr(>|t|)    
(Intercept)                  0.05427    0.29488   0.184  0.85416    
winnerwarriors              -0.32371    0.41363  -0.783  0.43481    
like_warriors               -0.86373    0.31342  -2.756  0.00641 ** 
winnerwarriors:like_warriors  1.81595    0.41592   4.366 2.05e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.899 on 196 degrees of freedom
Multiple R-squared:  0.09618, Adjusted R-squared:  0.08235 
F-statistic: 6.953 on 3 and 196 DF,  p-value: 0.0001805
```

Numerical x Categorical Interactions

Now you try it!

Exercise 2:
Intensity, determined by duration and prominent POS

Numerical x Numerical Interactions

When saying words in English, what is the effect of duration and pitch on the intensity (volume) of a syllable?

(Example 3)

Numerical x Numerical Interactions

```
> altM = lm(rintensity ~ rduration * rpitch, data = alt)
> summary(altM)
Call:
lm(formula = rintensity ~ rduration * rpitch, data = alt)
Residuals:
    Min      1Q  Median      3Q     Max 
-15.930  -3.471  -0.141   3.331  14.104 

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)    
(Intercept)        3.7567702  0.2670716  14.067  < 2e-16 ***
rduration          4.5849856  0.3449048  13.293  < 2e-16 ***
rpitch             0.0017614  0.0004086   4.311 1.94e-05 ***
rduration:rpitch -0.0014710  0.0005828  -2.524   0.0119 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.869 on 520 degrees of freedom
Multiple R-squared:  0.3272,	Adjusted R-squared:  0.3233 
F-statistic: 84.29 on 3 and 520 DF,  p-value: < 2.2e-16
```

Numerical x Numerical Interactions

Given an increase of one unit of duration, an increase of one unit of pitch reduces the intensity by 0.0015.

```
> altM = lm(rintensity ~ rduration * rpitch, data = alt)
> summary(altM)
Call:
lm(formula = rintensity ~ rduration * rpitch, data = alt)
Residuals:
    Min      1Q  Median      3Q     Max 
-15.930  -3.471  -0.141   3.331  14.104 

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)    
(Intercept)        3.7567702  0.2670716  14.067  < 2e-16 ***
rduration          4.5849856  0.3449048  13.293  < 2e-16 ***
rpitch             0.0017614  0.0004086   4.311 1.94e-05 ***
rduration:rpitch  -0.0014710  0.0005828  -2.524   0.0119 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.869 on 520 degrees of freedom
Multiple R-squared:  0.3272,	Adjusted R-squared:  0.3233 
F-statistic: 84.29 on 3 and 520 DF,  p-value: < 2.2e-16
```

Numerical x Numerical Interactions

```
                      Estimate Std. Error t value Pr(>|t|)
rduration:rpitch -0.0014710  0.0005828  -2.524   0.0119 *
```

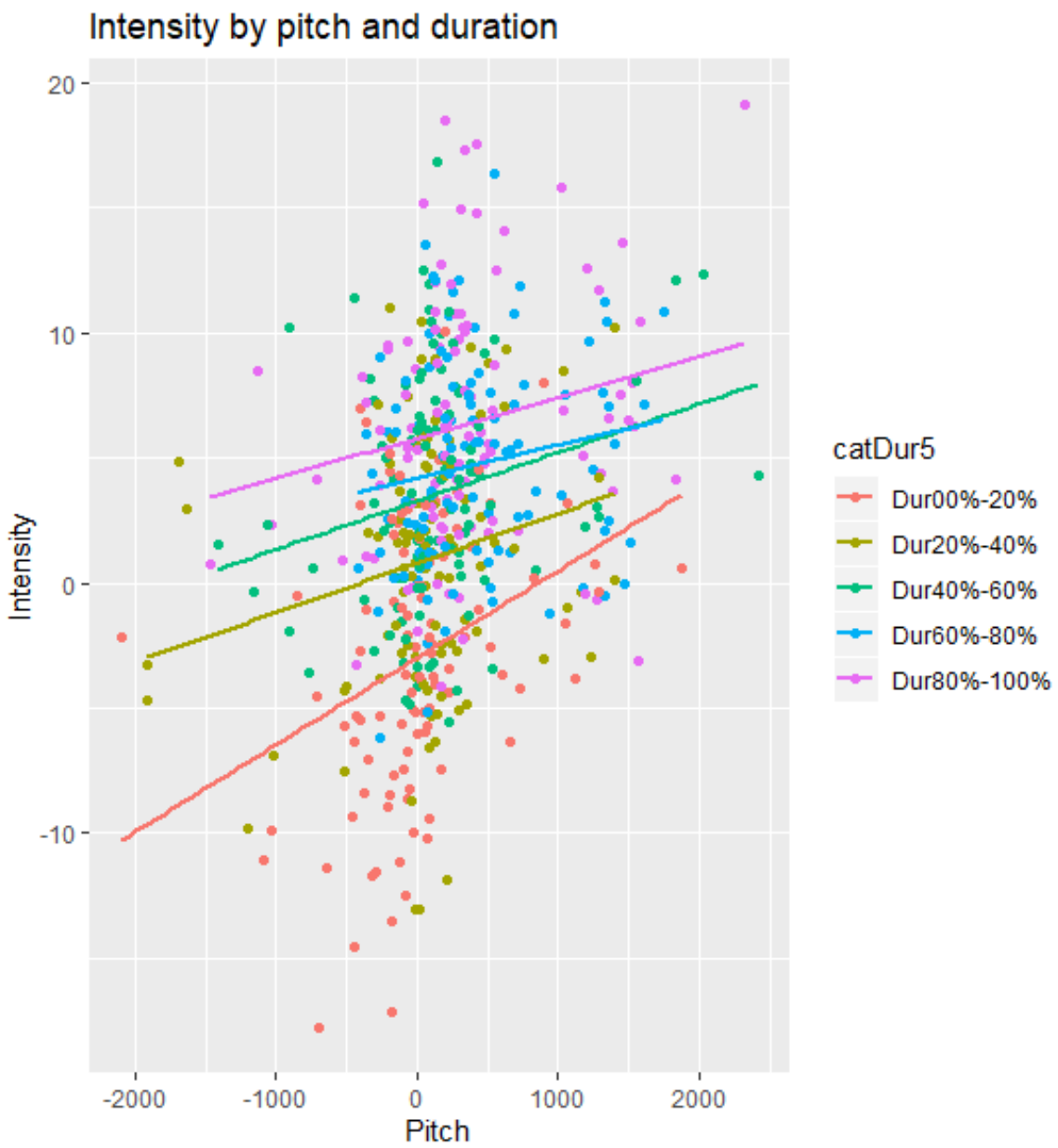
When duration changes by…	And pitch changes by…	Intensity
+1 unit	+1 unit	Decreases (-0.0015)
+1 unit	-1 unit	Increases (0.0015)
-1 unit	+1 unit	Increases (0.0015)
-1 unit	-1 unit	Decreases (-0.0015)

When only one of them increases, words get louder.
When both of them increase, words are not as loud (`dur` and `pitch` are mutually exclusive).
When both of them decrease, words are not as loud (e.g. function words)

Numerical x Numerical Interactions

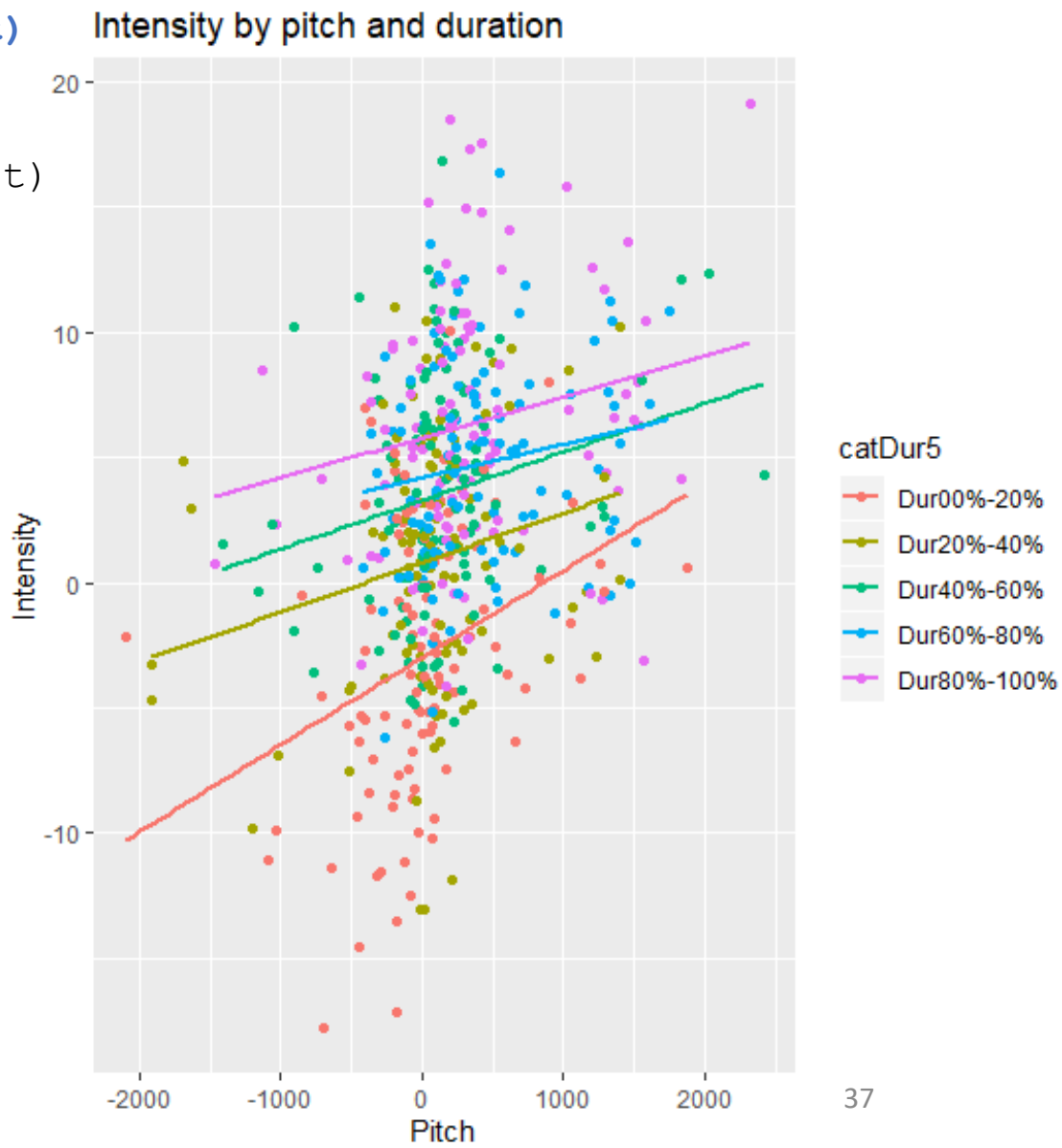
```
> summary(alt$rduration)

    Min.  1st Qu.   Median     Mean  3rd Qu.     Max. 
-1.85060 -0.80759 -0.39918 -0.33792  0.06508  1.63142 

> quantile(alt$rduration, prob = seq(0, 1, length = 6))

        0%        20%        40%        60%        80%       100% 
-1.8506000 -0.8710420 -0.5410771 -0.2321643  0.1453901  1.6314168 

> alt$catDur5[alt$rduration < -0.871] = "Dur00%-20%"
> alt$catDur5[alt$rduration >= -0.871 & alt$rduration < -0.541] = "Dur20%-40%"
> alt$catDur5[alt$rduration >= -0.541 & alt$rduration < -0.232] = "Dur40%-60%"
> alt$catDur5[alt$rduration >= -0.232 & alt$rduration <  0.145] = "Dur60%-80%"
> alt$catDur5[alt$rduration >= 0.145] = "Dur80%-100%"
```

Numerical x Numerical Interactions



Numerical x Numerical Interactions

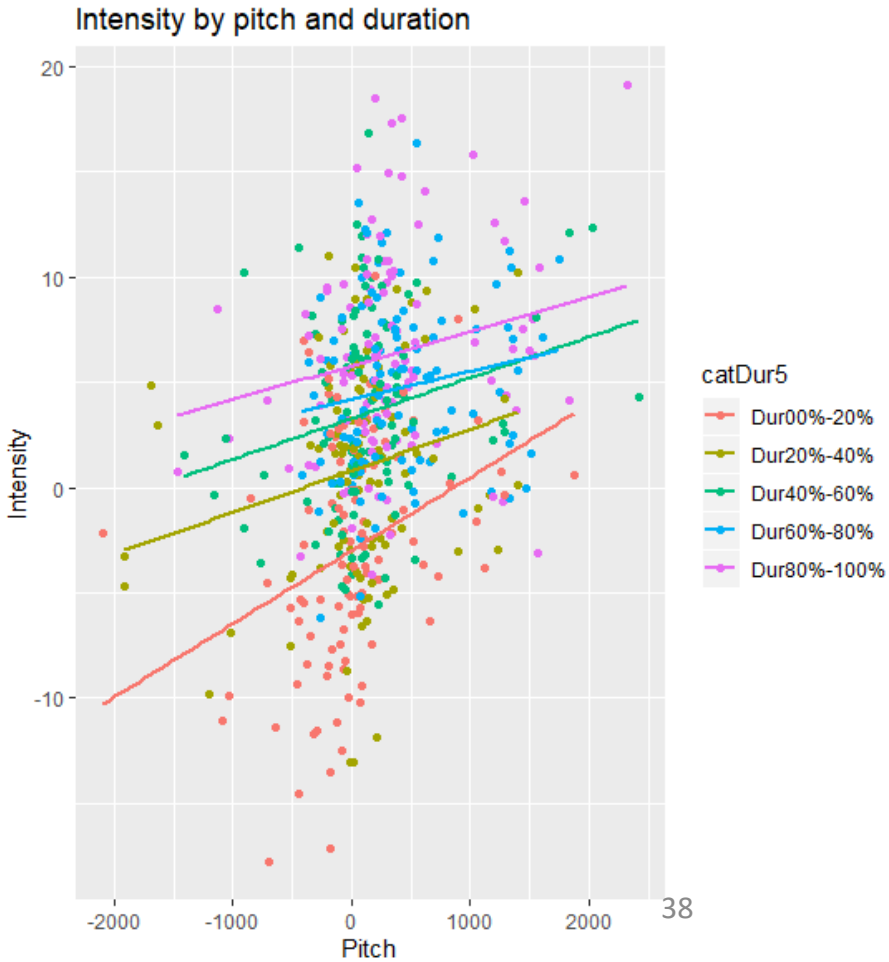
```
> altM = lm(rintensity ~ rduration * catDur5, data = alt)
> summary(altM)
Call:
lm(formula = rintensity ~ rduration * catDur5, data = alt)

                                Estimate Pr(>|t|)
(Intercept)                      12.617    2.67e-09 ***
rduration                        13.432      8.99e-14 ***
catDur5Dur20%-40%                -4.850      0.213192
catDur5Dur40%-60%                -5.480      0.066630 .
catDur5Dur60%-80%                -7.836      0.000268 ***
catDur5Dur80%-100%               -4.277      0.058669 .
rduration:catDur5Dur20%-40%      -3.942      0.417740
rduration:catDur5Dur40%-60%      -4.380      0.428657
rduration:catDur5Dur60%-80%     -11.501      0.009612 **
rduration:catDur5Dur80%-100%    -16.619      3.32e-14 ***
```



Numerical x Numerical Interactions

```
> altM = lm(rintensity ~ rduration * rpitch, data = alt)
Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)        3.7567702  0.2670716  14.067  < 2e-16 ***
rduration          4.5849856  0.3449048  13.293  < 2e-16 ***
rpitch             0.0017614  0.0004086   4.311 1.94e-05 ***
rduration:rpitch -0.0014710  0.0005828  -2.524   0.0119 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.869 on 520 degrees of freedom
Multiple R-squared:  0.3272,  Adjusted R-squared:  0.3233
F-statistic: 84.29 on 3 and 520 DF,  p-value: < 2.2e-16
```

There is a significant interaction between pitch and duration (t(520)=-2.5, p<0.05). In general, as pitch goes up, intensity goes up as well (t(520)=4.3, p<0.001). This effect is most visible in words of shorter duration (0-60 percentiles). However, in words of longer duration, pitch is less strongly correlated to intensity.



Model Selection for Linear Regression

Rolando Coto-Solano. LING10
Dartmouth College. February 2022

Parsimony and Model Selection

In science, we try to find an equilibrium between having theories that are **complete** (with all of the factors that influence the explanation) and that are **parsimonious** (as simple as posible).

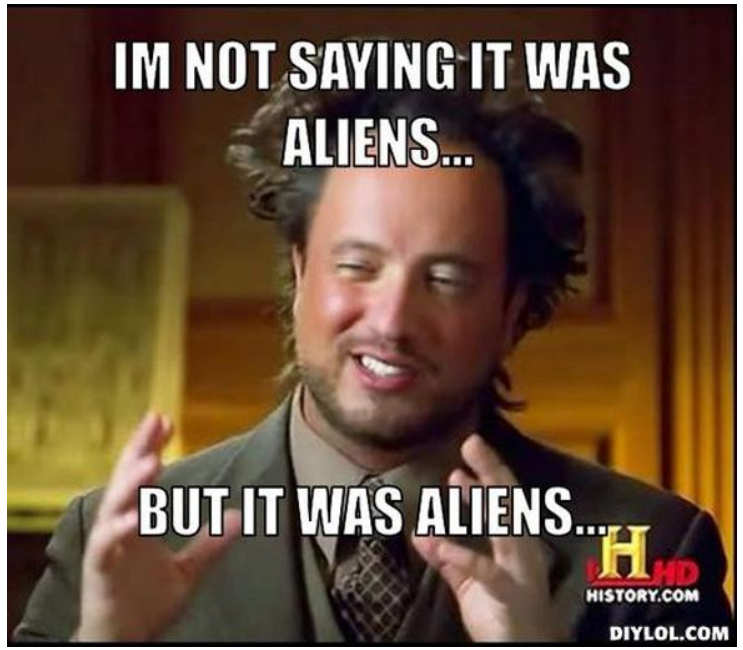
Parsimony and Model Selection

<u>Humans Built the Pyramids</u>

There are humans.
Humans frequently build huge and ornate tombs.
The pyramids are one example of this behavior.

<u>Aliens Built the Pyramids</u>

There is alien life.
Alien life has the technology to get to Earth.
Aliens are motivated to build pyramids.
Aliens left no unambiguous signs of their presence.
No scientists have discovered evidence for aliens.
Aliens covered their tracks by making the pyramids look like they were built by humans.



Parsimony and Model Selection

Occam's Razor (probably not from Ockham in the 1300s, but associated to him in the 1600s):

Non sunt multiplicanda entia sine necessitate

"Entities are not to be multiplied without necessity"

The principle itself is very old:

Aristotle: *We may assume the superiority, other things being equal, of the demonstration which derives from fewer postulates or hypotheses.*

Thomas Aquinas: *It is superfluous to suppose that what can be accounted for by a few principles has been produced by many*.

Pāṇini: Sutras/Grammar rules: *must have the minimum number of syllables. Not even one syllable should be extra or superfluous.* (Alpākṣaram)

(There must be a postulation about this from a Chinese philosopher. If you know it, please send me the reference!)

Parsimony and Model Selection

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It is a **heuristic**, not an absolute rule.

Parsimony and Model Selection

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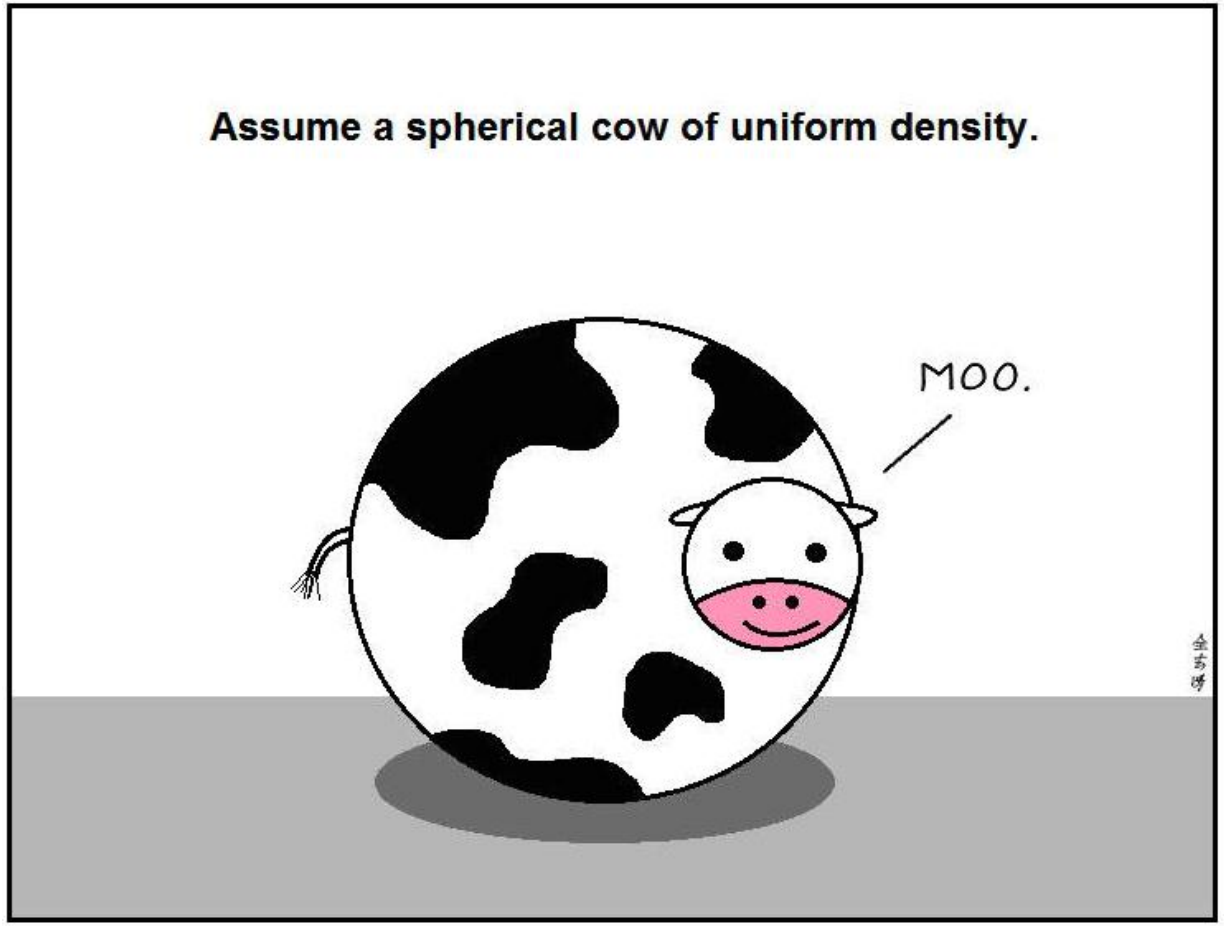
One reason we prefer simpler hypotheses is because they are **easier to falsify**, and therefore they are more easily testable by using the scientific method.

Parsimony and Model Selection



Parsimony means that we need to include as much as we need in our model to explain the phenomenon.
(But **think very carefully about what you're losing** when you eliminate a variable)

Parsimony and Model Selection

We have several tools to decide which predictors might be more or less useful for our models.

Parsimony and Model Selection

A tool to decide how much information is explained by a model is the **AIC, Akaike Information Criterion**

Parsimony and Model Selection

A tool to decide how much information is explained by a model is the **AIC, Akaike Information Criterion**

<u>Some explanations of what AIC is:</u>

If you've studied thermodynamics, AIC calculates *entropy*.

If you've studied information science, AIC minimizes the loss of information in a model.

If you're a mathematician, it's the *Kullback-Leibler Divergence Distance* (the distance between the current model and an ideal model).

AIC is a measure of how well a model fits a certain dataset, and whether it would fit other datasets just as well.

Parsimony and Model Selection

A tool to decide how much information is explained by a model is the **AIC**.

Punishment for model complexity; adding predictors will always improve the prediction just by chance

$$AIC = \underbrace{n\left(log\frac{SSE}{n}\right)}_{\text{Likelihood of the observed variation, given the calculated coefficients.}} + \overbrace{2(p+1)}$$

Honestly, the only intuitive explanation for Akaike that I've been able to understand is this one: https://www.r-bloggers.com/how-do-i-interpret-the-aic/

Example: German Incomplete Neutralization

In German, voiced stops become voiceless at the end of a word.

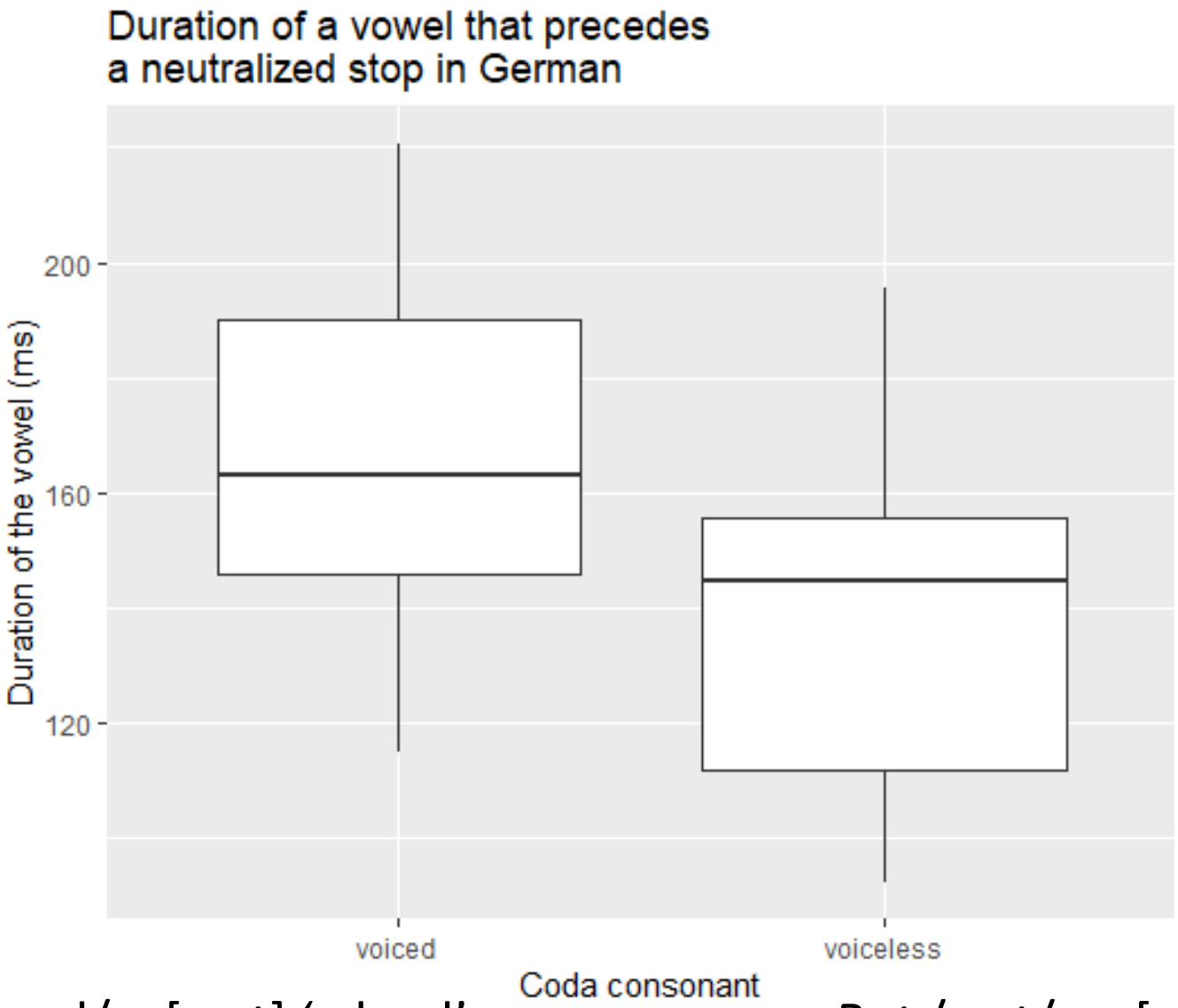
a. *Rat* /ʁa:t/ > [ʁa:t] ‘council’, *Räte* [ʁɛ:tɐ] ‘councils’
b. *Rad* /ʁa:d/ > [ʁa:t] ‘wheel’, *Räder* [ʁɛ:dɐ] ‘wheels’

Because of this change, the word *Rad* ‘wheel’ [ʁa:t] sounds the same as the word *Rat* ‘council’ [ʁa:t]. The ‘t’ and ‘d’ are no longer different. This process is called **neutralization**.

People used to think that the 't' in *Rat* was identical to the 't' in *Rad*. However, recent research has shown that they are different. One of their differences is that the vowels that precede them have different durations.

How can we predict the duration of a vowel that precedes a neutralized stop in German?

Example: German Incomplete Neutralization



Rad /ʁa:d/ > [ʁ**a:**t] ‘wheel’

Rat /ʁa:t/ > [ʁ**a:**t] ‘council’

Example: German Incomplete Neutralization

```
> oneVarNeutral = lm(item_vowel_dur ~ voicing, data = neutral)
> summary(oneVarNeutral)
Call:
lm(formula = item_vowel_dur ~ voicing, data = neutral)

Residuals:
    Min      1Q  Median      3Q     Max 
-50.923 -22.654   2.487  19.361  59.101 

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)    
(Intercept)       165.943      1.503  110.41   <2e-16 ***
voicingvoiceless  -29.454      2.095  -14.06   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 27.55 on 690 degrees of freedom
Multiple R-squared:  0.2226,	Adjusted R-squared:  0.2215 
F-statistic: 197.6 on 1 and 690 DF,  p-value: < 2.2e-16
```

Example: German Incomplete Neutralization

```
> AIC(oneVarNeutral)
[1] 6557.163
```

The AIC for the model with one variable.

```
> summary(oneVarNeutral)$r.squared
[1] 0.222597
```

The R-Squared of the model with one variable.

```
> summary(oneVarNeutral)$adj.r.squared
[1] 0.2214703
```

The "adjusted" R-Squared makes a correction for having more than one variable.

Example: German Incomplete Neutralization

```
lm(formula = item_vowel_dur ~ voicing + utterancelength + accent_type +
    prosodic_boundary + place + subject, data = neutral)

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)          173.60642   12.08236  14.369  < 2e-16 ***
voicingvoiceless     -29.56696    2.05566 -14.383  < 2e-16 ***
utterancelength       -3.35387    5.77809  -0.580    0.562
accent_typenuclear     4.98042   11.42721   0.436    0.663
accent_typeprenuclear  1.74161    8.65367   0.201    0.841
prosodic_boundaryyes  -1.18515    7.76592  -0.153    0.879
placelabial          -14.06850    2.59890  -5.413 8.58e-08 ***
placevelar            -1.75743    2.55265  -0.688    0.491
subject                0.07791    0.23285   0.335    0.738
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 26.96 on 683 degrees of freedom
Multiple R-squared:  0.2633,  Adjusted R-squared:  0.2547
F-statistic: 30.51 on 8 and 683 DF,  p-value: < 2.2e-16
```

Example: German Incomplete Neutralization

Model with one variable

```
> AIC(oneVarNeutral)
[1] 6557.163
```

```
> summary(oneVarNeutral)$
                r.squared
[1] 0.222597
```

```
> summary(oneVarNeutral)$
            adj.r.squared
[1] 0.2214703
```

Model with six variables

```
> AIC(maxNeutral)
[1] 6533.948
```

```
> summary(maxNeutral)$
             r.squared
[1] 0.2633006
```

```
> summary(maxNeutral)$
         adj.r.squared
[1] 0.2546716
```

Example: German Incomplete Neutralization

```
> stepAIC(lm(neutral$item_vowel_dur~1),
+         neutral$item_vowel_dur ~ neutral$voicing +
neutral$utterancelength + neutral$accent_type +
neutral$prosodic_boundary + neutral$place + neutral$subject,
+         direction="forward")
Start:  AIC=4763.59
neutral$item_vowel_dur ~ 1

                          Df Sum of Sq    RSS    AIC
+ neutral$voicing          1    149958 523717 4591.4
+ neutral$place            2     26534 647141 4739.8
<none>                                 673675 4763.6
+ neutral$subject          1       155 673520 4765.4
+ neutral$prosodic_boundary 1        1 673675 4765.6
+ neutral$utterancelength  1         0 673675 4765.6
+ neutral$accent_type      2       140 673535 4767.5
```

Example: German Incomplete Neutralization

```
Step:  AIC=4591.35
neutral$item_vowel_dur ~ neutral$voicing
                          Df Sum of Sq    RSS    AIC
+ neutral$place            2   26723.1 496994 4559.1
<none>                                 523717 4591.4
+ neutral$prosodic_boundary  1     123.3 523594 4593.2
+ neutral$utterancelength    1      25.1 523692 4593.3
+ neutral$subject            1      13.0 523704 4593.3
+ neutral$accent_type        2     247.8 523469 4595.0

Step:  AIC=4559.11
neutral$item_vowel_dur ~ neutral$voicing + neutral$place
                          Df Sum of Sq    RSS    AIC
<none>                                 496994 4559.1
+ neutral$prosodic_boundary  1    281.21 496713 4560.7
+ neutral$subject            1     40.75 496953 4561.1
+ neutral$utterancelength    1     19.14 496975 4561.1
+ neutral$accent_type        2    392.32 496602 4562.6
```

Example: German Incomplete Neutralization

```
Call:
lm(formula = neutral$item_vowel_dur ~ neutral$voicing + neutral$place)

Coefficients:
             (Intercept)  neutral$voicingvoiceless       neutral$placelabial
                 171.393                   -29.480                   -13.993
      neutral$placevelar
                  -1.736
```

Example: German Incomplete Neutralization

```
> minNeutral = lm(item_vowel_dur ~ voicing + place, data = neutral)
> summary(minNeutral)
Call:
lm(formula = item_vowel_dur ~ voicing + place, data = neutral)
Residuals:
    Min      1Q  Median      3Q     Max
-55.043 -25.820  -1.203  22.607  55.413
Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)       171.393      2.196  78.033  < 2e-16 ***
voicingvoiceless  -29.480      2.045 -14.417  < 2e-16 ***
placelabial       -13.993      2.588  -5.407 8.84e-08 ***
placevelar         -1.736      2.542  -0.683    0.495
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 26.88 on 688 degrees of freedom
Multiple R-squared:  0.2623,  Adjusted R-squared:  0.259
F-statistic: 81.53 on 3 and 688 DF,  p-value: < 2.2e-16
```

Example: German Incomplete Neutralization

Model with one variable

```
> AIC(oneVarNeutral)
[1] 6557.163
```

```
> summary(oneVarNeutral)$
                r.squared
[1] 0.222597
```

```
> summary(oneVarNeutral)$
            adj.r.squared
[1] 0.2214703
```

Model with six variables

```
> AIC(maxNeutral)
[1] 6533.948
```

```
> summary(maxNeutral)$
             r.squared
[1] 0.2633006
```

```
> summary(maxNeutral)$
         adj.r.squared
[1] 0.2546716
```

Model with two variables

```
> AIC(minNeutral)
[1] 6524.92
```

```
> summary(minNeutral)$
             r.squared
[1] 0.2622645
```

```
> summary(minNeutral)$
         adj.r.squared
[1] 0.2590477
```

Summary so far

The AIC is a measure of how well the variables explain the data. We always prefer the lower value of AIC.

You can use instructions like `stepAIC` to get the optimal model for your data.

Exercise 4: ELP / AIC

In English, the reaction time to a word can be determined by its word frequency, its length (in letters), and the part of speech (adjectives, nouns, verbs).

But, do we really need all three predictors? Run the example and tell us what variables we actually need.